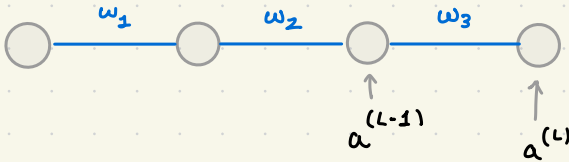


BACK PROPAGATION

$$C(w_1 b_1, w_2 b_2, w_3 b_3)$$

How sensitive is the **cost function** to these variables?

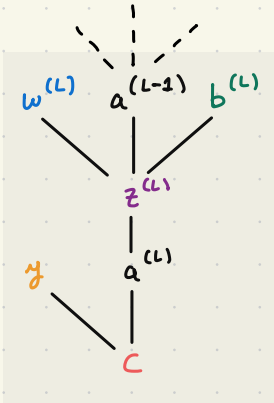


$y :=$ DESIRED OUTPUT

$$\Rightarrow C_o(\dots) = (a^{(L)} - y)^2 \quad \leftarrow \text{For ONE Training Example!}$$

$$a^{(L)} := \sigma(\underbrace{w^{(L)} a^{(L-1)} + b^{(L)}}_{= z^{(L)}}) \quad \leftarrow \sigma: \text{any non-linear } f$$

$$\Rightarrow a^{(L)} = \sigma(z^{(L)})$$



First Goal: $\frac{\partial C_o}{\partial w^{(L)}}$

How sensitive is the cost to the weight?

Chain Rule

$$\frac{\partial C_o}{\partial w^{(L)}} = \frac{\partial z^{(L)}}{\partial w^{(L)}} \cdot \frac{\partial a^{(L)}}{\partial z^{(L)}} \cdot \frac{\partial C_o}{\partial a^{(L)}}$$

$$\cdot \frac{\partial z^{(L)}}{\partial w^{(L)}} = (w^{(L)} a^{(L-1)} + b^{(L)})' = a^{(L-1)}$$

$$\cdot \frac{\partial a^{(L)}}{\partial z^{(L)}} = \sigma'(z^{(L)})$$

$$\cdot \frac{\partial C_o}{\partial a^{(L)}} = ((a^{(L)} - y)^2)' = 2(a^{(L)} - y)$$

$$\Rightarrow \frac{\partial C_o}{\partial w^{(L)}} = a^{(L-1)} \sigma'(z^{(L)}) \cdot 2(a^{(L)} - y) \rightarrow \text{Cost of ONE Training Example!}$$

$\Rightarrow$ Total Cost: $\frac{\partial C}{\partial w^{(L)}} = \frac{1}{n} \sum_{k=0}^{n-1} \frac{\partial C_k}{\partial w^{(L)}} \rightarrow$ Average of ALL Train. Examples

But This is just one component of the **gradient vector**!

$$\nabla C = \begin{bmatrix} \frac{\partial C}{\partial w^{(1)}} \\ \frac{\partial C}{\partial b^{(1)}} \\ \vdots \\ \frac{\partial C}{\partial w^{(L)}} \\ \frac{\partial C}{\partial b^{(L)}} \end{bmatrix}$$

$\Rightarrow$ we need to find the sensitivity to the **bias**

Same process

- Second goal: $\frac{\partial C}{\partial b^{(L)}} = [\dots] = 1 \cdot \sigma'(z^{(L)}) \cdot 2 \cdot (a^{(L)} - y)$

- Third goal: $\frac{\partial C}{\partial a^{(L-1)}}$ $\leftarrow$ How sensitive it is to the previous layer?

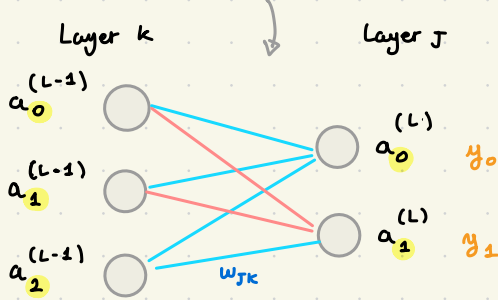
Idea of BACKPROPAGATION

$$\frac{\partial C}{\partial a^{(L-1)}} = \frac{\partial z^{(L)}}{\partial a^{(L-1)}} \cdot \underbrace{\frac{\partial a^{(L)}}{\partial z^{(L)}} \cdot \frac{\partial C}{\partial a^{(L)}}}_{\text{DONE}}$$

$$z^{(L)} = w^{(L)} a^{(L-1)} + b^{(L)} \Rightarrow \frac{\partial z^{(L)}}{\partial a^{(L-1)}} = w^{(L)}$$

$$\frac{\partial C}{\partial a^{(L-1)}} = w^{(L)} \sigma'(z^{(L)}) \cdot 2 \cdot (a^{(L)} - y)$$

- What happens on multiple neurons?



$$C_0 = \sum_{j=0}^{n_L-1} (a_j^{(L)} - y_j)^2$$

$$\Rightarrow z_j^{(L)} = w_{j0}^{(L)} a_0^{(L-1)} + w_{j1}^{(L)} a_1^{(L-1)} + w_{j2}^{(L)} a_2^{(L-1)} + b_j^{(L)}, \quad a_j^{(L)} = \sigma(z_j^{(L)})$$

We now apply the same chain rule as before

$$\bullet \frac{\partial C_0}{\partial w_{jk}^{(L)}} = \frac{\partial z_j^{(L)}}{\partial w_{jk}^{(L)}} \frac{\partial a_j^{(L)}}{\partial z_j^{(L)}} \cdot \frac{\partial C_0}{\partial a_j^{(L)}} = [\dots]$$

$$\bullet \frac{\partial C_0}{\partial a_k^{(L-1)}} = \underbrace{\sum_{j=0}^{n_L-1} \frac{\partial z_j^{(L)}}{\partial a_k^{(L-1)}} \frac{\partial a_j^{(L)}}{\partial z_j^{(L)}} \cdot \frac{\partial C_0}{\partial a_j^{(L)}}}_{\text{Sum over layer } L}$$



Each neuron influences
ALL the neurons after
 $\Rightarrow$ ADD them up